

Speak Like A Mathematician:

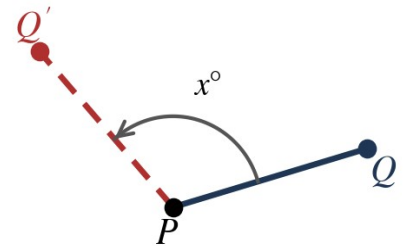
rotation:

center of rotation:

angle of rotation:

rotational symmetry:

center of symmetry:



Coordinate Rules for Rotations About the Origin

All rotations here are _____ (counterclockwise unless you are told otherwise).

90°: $(a, b) \rightarrow (\text{_____, _____})$ **180°:** $(a, b) \rightarrow (\text{_____, _____})$

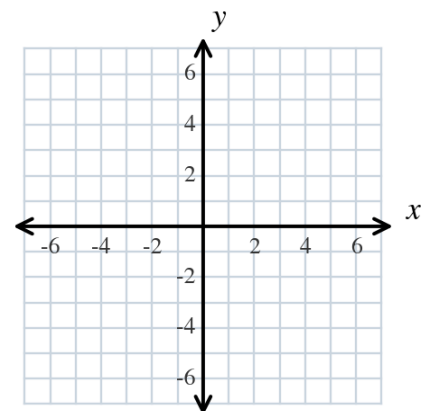
270°: $(a, b) \rightarrow (\text{_____, _____})$

Rotation Postulate: a rotation is a _____.

Example 1 - Rotate 90° About the Origin

Graph quadrilateral WXYZ with $W(1, 2)$, $X(4, 3)$, $Y(5, -1)$, and $Z(2, -2)$ and its image after a 90° rotation about the origin.

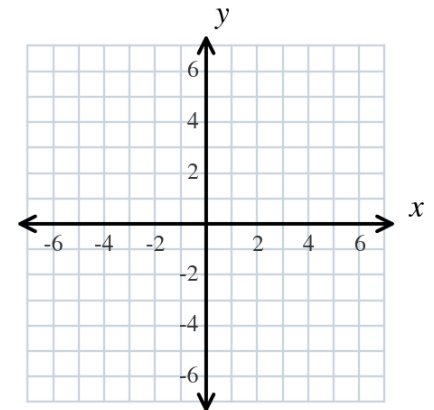
- State the rule you will use.
- Find the coordinates of the image.
- Is $WXYZ \cong W'X'Y'Z'$? Explain.



Example 2 - Rotate 180° About the Origin

Graph $\triangle PQR$ with $P(-2, 1)$, $Q(1, 4)$, and $R(3, 0)$ and its image after a 180° rotation about the origin.

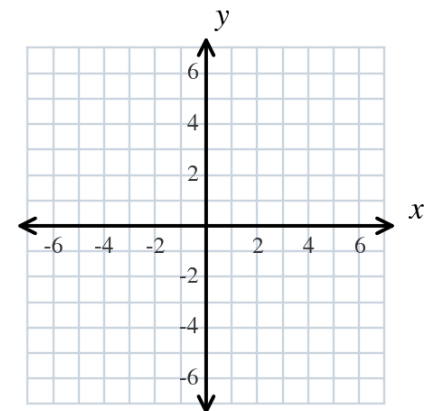
- Find the coordinates of the image.
- Why does it not matter whether the 180° turn is clockwise or counterclockwise?



Example 3 - Rotate 270° About the Origin

Graph $\overline{ST}$ with $S(-3, 2)$ and $T(-1, 5)$ and its image after a 270° rotation about the origin.

- State the rule you will use.
- Find the coordinates of the image.



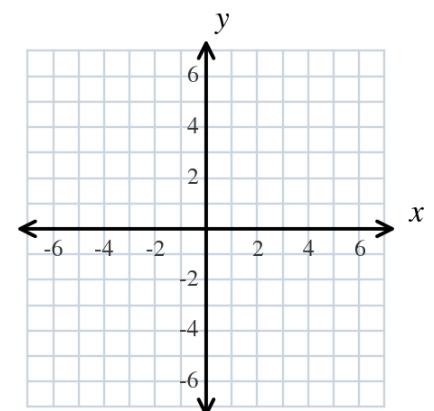
Example 4 - Reflection, Then Rotation

Graph $\overline{AB}$ with $A(2, -4)$ and $B(4, -1)$ and its image after the composition.

Reflection: in the y-axis

Rotation: 90° about the origin

- Find the coordinates after the reflection.
- Find the coordinates after the rotation.
- Does the order matter? Explain.



Rotational Symmetry

A figure has rotational symmetry when a rotation of _____ or less about its center maps the figure

_____.

That center point is the center of _____. The rotation may be clockwise or counterclockwise.

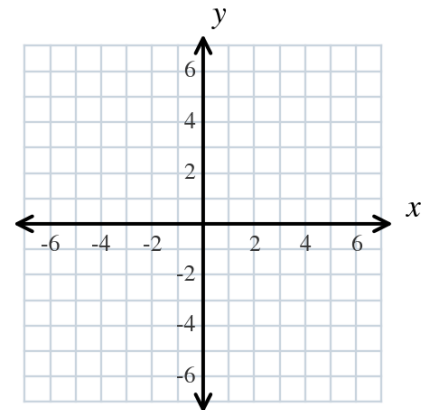
Example 5 - Translation, Then Rotation

Graph $\triangle CDE$ with $C(-6, 2)$, $D(-4, 5)$, and $E(-2, 3)$ and its image after the composition.

Translation: $(x, y) \rightarrow (x + 2, y - 6)$

Rotation: 180° about the origin

- Find the coordinates after the translation.
- Find the coordinates after the rotation.



Example 6 - Rotational Symmetry

Does the figure have rotational symmetry? If so, describe every rotation of 180° or less that maps it onto itself.

- regular hexagon
- rectangle (not a square)
- regular pentagon
- isosceles trapezoid

